Installation and Commissioning Instructions

Application and Installation (A&I)

MotivLine Series 4000 C03

with

- ECU 7/EMU 7
- SAM
- POM

E532256/01E

Power. Passion. Partnership.

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

1 Safety

1.1 General conditions

General

In addition to the instructions in this publication, the applicable country-specific legislation and other compulsory regulations regarding accident prevention must be observed. This state-of-the-art system has been designed to meet all applicable laws and regulations. The system may nevertheless present a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Modifications or conversions
- Non-compliance with the Safety Instructions

Correct use

The system is intended solely for use in accordance with contractual agreements and the purpose envisaged for it on delivery. Any other use is considered improper use. The system manufacturer accepts no liability whatsoever for resultant damage or injury in such case. The system manufacturer will not be liable for any losses and/or damages caused by improper use or actions resulting from personnel while under the influence of alcohol, medication or any other mind altering substances. The responsibility is borne by the user alone.

Correct use also includes observation of and compliance with the maintenance schedule.

Liability exclusion

In the case of nonobservance of the specifications and information contained in this guideline, the manufacturer shall be released from liability and warranty obligations. No guarantee is provided that information in this guideline covers all requirements with regard to design of the plant or difficulties with installation. For this reason, MTU shall not entertain warranty claims, in particular, claims related to configuration and the installation of bought-in products. Changes that contribute to technical progress can also be made without prior notice.

Due to their diversity, locally valid laws, ordinances and specifications cannot be covered in this document, although they must be observed (see below for examples).

Numerical values in this guideline serve to clarify magnitude and must be regarded as examples.

Proposals included in this guideline for preventing dangerous operational statuses are necessarily of a general nature because, with regard to each partial system used, only the respective OEM (Original Equipment Manufacturer) is familiar with all details of his equipment. The examples do not guarantee prevention of a dangerous operational status in all circumstances.

Each system installation requires an MTU release following the installation inspection (EPQ).

Operational reliability, dependability and a long service life are also influenced by observance of the specified maintenance tasks. During the planning phase and installation of the plant, therefore, care must be taken to ensure easy access for servicing, maintenance and repair tasks.

Safety notifications used in this publication

Safety notifications (DANGER, WARNING, and CAUTION) are used in this manual to alert the owner and/or operator to special instructions concerning a particular procedure that may be hazardous if performed incorrectly.

These safety notifications alone cannot eliminate the hazards that they signal. Strict compliance with these special instructions and common sense operation are critical accident prevention measures. All safety notifications found on the equipment must be observed. Ensure that safety-related labels are legible and not obstructed by dirt, grease or other equipment.

DANGER	In the event of immediate danger **Consequences: Death or serious injury.** • Preventive measures

WARNING	In the event of possibly dangerous situations. **Consequences: Death or serious injury.** • Preventive measures
CAUTION	In the event of dangerous situations. **Consequences: Slight injury or material damage.** • Preventive measures
NOTICE	Meaning of NOTICE • Notice is used throughout the publication to communicate a requirement, recommendation or fact relating to proper use, transport, installation, operation, or maintenance of the equipment but is not hazard related
NOTICE	Manufacturers' recommendations • All manufacturers' safety notifications are to be followed regardless of format.

Read and become acquainted with all safety notifications and their meaning before working with this system. Contact an authorized service outlet if clarification is needed.

The manufacturer cannot anticipate every possible circumstance that might involve a potential hazard. The warnings in this publication and on the product are, therefore, not all inclusive. If a tool, procedure, work method or operating technique that is not specifically recommended by the manufacturer is used, you must satisfy yourself that it is safe for you and for others. Ensure the product will not be damaged or made unsafe by the transport, installation, commissioning, operation, lubrication, maintenance, repair or storage procedures that you choose.

The information, specifications and illustrations in this manual are on the basis of information available at the time it was written. The information, specifications and illustrations can change at any time. These changes can affect the service given to the product. Obtain the complete and most current information before you start any work on the system.

Modifications or conversions

Unauthorized modifications to the system represent a safety risk.

MTU will accept no liability or warranty claims for any damage caused by unauthorized modifications, third-party add-on devices and/or replacement parts. In the event that damage is caused by the use of non-genuine replacement parts, no liability or warranty claims through the system manufacturer will be accepted.

NOTICE	Manufacturers' safety notifications • All manufacturers' safety notifications are to be followed. • Unauthorized modifications, third-party add-on devices and replacement parts may compromise the integrity of the equipment, and may present a safety risk.
NOTICE	Replacement safety labels • Please contact an authorized service outlet to determine how to obtain replacement labels for parts that have safety labels.

Spare parts

Only genuine MTU spare parts are allowed to be used to replace components or assemblies. The system manufacturer accepts no liability whatsoever for damage or injury resulting from the use of other spare parts and the warranty shall be voided in such case.

Reworking components

Repair must be carried out in workshops authorized by MTU.

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1.2 Personnel and organizational requirements

Publication notice

For the remainder of this publication:

- The term “work”, “working” or “all work” is used in reference to work done on the system and is defined as one or more of the following actions: transport, installation, commissioning, operation, maintenance, repair and/or storage.
- The term “certified personnel” refers to a certified technician who has successfully completed training at MTU's certified training center and who has also been instructed on the system work by means of this publication, with special attention paid to the safety notifications. MTU's factory certified technician training classes are available in various locations to meet worldwide needs.
- The term “operator” or “third party user” is the person involved with the transport, installation, commissioning, operation, maintenance and/or repair that does not require him/her to be an MTU certified technician, but who must read and have full understanding of this publication, especially the safety notifications, before proceeding with any operation.
- The term “owner” or “person responsible for the engine operation” is the person who either owns the system or facility within which it resides. He/she generally schedules or directs the work being done on or with the system.
- The term “third-party” or “add-on” generally refers to items not designed by or purchased from MTU; whereas, “genuine” or “factory” means it was designed by or purchased from MTU.

Personnel and organizational requirements

This publication must be provided to all certified personnel involved in any or all work being done on the system and should be located in the vicinity of the system to be accessible at all times.

In general, operational checks can be performed by non-certified personnel following the published instructions. All other work on the system must be carried out by properly certified personnel.

All personnel working on the system must ensure that all safety notifications and informational messages are read and understood before any work is performed on the system.

In addition, observe the following:

- All work on the system shall be carried out by trained and qualified personnel only.
- The specified legal minimum age must be observed.
- The operator must specify the responsibilities of the operating, maintenance and repair personnel.
- Owners are responsible for instructing personnel who only occasionally work on the system, prior to requesting the work be performed. This personnel must be instructed repeatedly.

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1.3 Safety requirements in general

Working on an engine

Mechanical equipment can cause bodily harm and pose life-threatening danger when improperly transported, installed, commissioned, operated, maintained or repaired. Do not work on this engine unless you have read and understood the instructions and safety notifications contained in this publication. Failure to follow the instructions or heed the safety notifications could result in injury or death.

NOTICE

Persons performing work on equipment

- Every person performing work on the equipment should read and become acquainted with all cautions and symbols before operating or repairing this product.

NOTICE

Permission for working on the equipment

- Always obtain permission from the person in charge before commencing work on the equipment.

NOTICE

Additional safety standards

- Additional national safety standards exist. The manufacturer recommends owners, operators and certified technicians become familiar with national safety standards, where available.
- Operators should refer to the safety requirements for operators in this publication.

General safety requirements for operators

In addition, read and understand the following:

- Never carry out maintenance and/or repair work with the engine running, unless specified in the procedure and only in accordance with the applicable safety messages.
- Never work on the engine or other engine components without following proper lifting requirements.
- Never attempt to rectify faults or carry out repairs without the necessary certification and required special tool.
- Strictly adhere to the maintenance and repair schedule.
- Perform tasks as described.
- Use proper, calibrated tools and equipment.
- Always ensure that nobody is standing in the danger zone prior to barring over the engine.
- After completing work on the engine, check that all protective devices and safety guards have been installed. Remove all tools and loose objects from the engine vicinity.
- Ensure persons not involved in the work on the engine keep clear.
- Observe special cleanliness when conducting maintenance and repair work on the engine.
- For the identification and layout of the spare parts during maintenance or repair work, take photos or use the spare parts catalog.
- Regularly practice procedures to be prepared in cases of emergency.
- Be familiar with the engine controls and displays.
- Observe the displays and monitoring units with regard to present operating status and warning or alarm messages.

The following steps must be taken if a malfunction of the system occurs or is displayed:

1. Acknowledge the message and take appropriate action.
2. Notify the supervisory personnel in charge.
3. Carry out emergency actions (e.g. emergency engine stop), if required.

Personal protection

To reduce the risk of injury:

- Never work on the system while under the influence of alcohol, medication or any other mind-altering substances.
- Use the necessary protective equipment for the given work to be done.
- Wear protective clothing and approved safety shoes for all work.
- Use face, eye, ear and nose protection.

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